

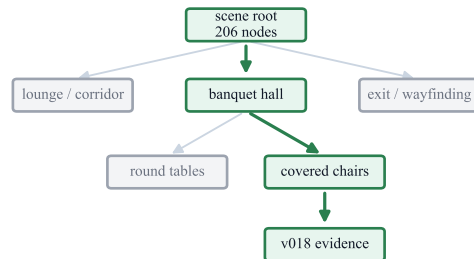
A Posed RGB-D capture



B Semantic ViewJSON record



C Deterministic hierarchy assembly



five scenes: 97 views | 1,066 items | 0 provider construction tokens

PER-QUERY RETRIEVAL

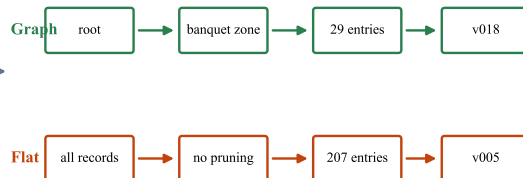
D Natural-language query



target covered chairs

attribute white

E Top-down pruning; identical scorer



views: 3 vs 20 | estimated context: 555 vs 3,110 tokens

F Grounded evidence



graph hit@1: yes | 2.07x speedup | 82.2% fewer est. tokens